

How do input modes affect politeness?

Chanmi Lee

chanmi.lee@stud.uni-regensburg.de
University of Regensburg
Regensburg, Germany

Fabian Aleksic

fabian.aleksic@stud.uni-regensburg.de
University of Regensburg
Regensburg, Germany

Felix Braun

felix1.braun@stud.uni-regensburg.de
University of Regensburg
Regensburg, Germany

Ema Strinic

ema.strinic@stud.uni-regensburg.de
University of Regensburg
Regensburg, Germany

Abstract

Large language model (LLM) systems are increasingly used through devices with very different input constraints, yet there is little research about how a device's form factor influences the way the users addresses them. In this paper, we examine whether desktop and mobile input conditions are associated with differences in the use of politeness expressions in task-oriented chatbot interaction. We analyze an annotated dataset of cooking-task conversations in which participants interact with a chatbot while following recipes. The final dataset contains 22 participants (11 desktop, 11 mobile), 132 recipe-specific conversations, and 1,406 annotated user utterances. We first compare the two groups on self-reported typing speed and basic conversational behavior, and then test differences in politeness-marker use. Our main politeness outcome combines utterances containing *please*, thanking, apology, compliment/face-enhancing feedback, or acknowledgment of response. Descriptively, desktop participants produce more utterances and show a higher politeness-marker rate than mobile participants. In a mixed-effects Poisson model controlling for conversation length, recipe, assignment order, and repeated observations within participants, mobile users show a lower politeness-marker rate than desktop users (IRR = 0.72, 95% CI [0.52, 1.00], $p = .049$). These findings suggest that input constraints shape how politely users interact with LLM-based systems.

Keywords

Politeness, Chatbots, Input device, Large language models (LLMs), User study

ACM Reference Format:

Chanmi Lee, Felix Braun, Fabian Aleksic, and Ema Strinic. 2026. How do input modes affect politeness?. In *Proceedings of (Interacting With AI Systems)*. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Interacting With AI Systems, Regensburg, Deutschland

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

Large Language Models have achieved a huge boom over the past years, and are being included in more products every day [8]. They are starting to supplant many traditional app-based approaches, and quickly becoming a major entry-point into people's computing devices [6]. This results in a shift towards using natural language as an input method, rather than just using buttons and menus [7]. Users have diverse ways of interacting with their devices, such as full-size hardware keyboards on notebooks, to touchscreen keyboards of varying sizes, all the way to speech input on smartwatches or via headphones [3]. These different input modes already require some optimizations to enable applications that require input to work well, such as predefined answers for incoming messages on smartwatches [2].

With the relevance of politeness already demonstrated [4], we build on this existing research and investigate the effect different input methods have on this. We believe that on top of an individual's baseline politeness (resulting both from character as well as mental model [14] there's a bias introduced by the throughput of the input method used to communicate with the chatbot.

Input methods such as smartwatch displays require considerably longer time to input the same number of words [15], making it more expensive to use pleasantries and other formalities not directly required by the query. Using smaller screens for typing also results in quicker fatigue and less comfort [9]. We suspect that this results in a more direct, but also less polite interaction, compounded by (potential) frustration at the difficulty in getting the point across.

A very common scenario for using smart assistants is the kitchen, where people want help with following recipes, and asking questions about what they need to do next. We're going to be simulating this scenario for our study.

2 Related work

There is extensive research into the topics of politeness, linguistic style and how different personalities affect the interaction with chatbots, both from a technical as well as an ethical standpoint [11]. On the technical level, several studies show that the phrasing and structure of a query—of which politeness strategies are a part—can influence not only the quality of chatbot responses but also their generation speed, since variations in prompt length and complexity directly affect computational load and response latency [1, 16].

One study that's of particular relevance to us is the Study by Elswiler et al. [4]. This study shows that politeness systematically

affects not just the response length, but also how dense information returned is. This is of increasing relevance due to the impacts on power consumption and required computation when running LLM models locally in embedded devices. Also, when chatbots are used as primary interaction methods with devices and apps, the length and density of information is especially important, since the display size can vary [17], and audio responses take time to be played back [12]. Limitations in communication bandwidth can go both ways, and costs are incurred when the transmitted content is inefficient, mainly in terms of the time required for an interaction [10]. The paper by *Martina Ziefle* holds that [17] text size and information density need to be carefully considered to keep users well oriented.

3 Study Design

Our study specifically focuses on *device-form-factor differences* within typed input. We chose to have both groups use typing as the input method, out of concern for possible bias due to different mental models and habits between spoken and written language, affecting our results.

3.1 Study type

We conducted an online-study, where participants are randomly assigned to one of two conditions, mobile or desktop. To avoid fatigue-effects, our study was set up as between-groups, so every participant only performed the task on a single device. We supervised the study via video call to ensure the participants adhere to our guidelines and use appropriate devices.

3.2 Ecological Validity

A kitchen is oftentimes a very loud environment, so typed input is likely to be used in a real setting as well. We also believe that this experiment scales to different topics and settings, since there's nothing about our hypothesis or test that's exclusive to kitchen settings. However, using this scenario is helpful, because it allows us to compare our findings more closely to the existing study by Elweiler et al. [4].

4 Research question

Research into how the input methods affect users' behavior towards LLMs remains scarce, specifically with the context of politeness. We suspect that the input device has an effect on the usage of politeness markers, due to the additional effort required to add politeness phrases. This leads us to the following research question to address this gap:

RQ1: Does the input device impact the usage of politeness expressions in conversations with an LLM-based assistant?

5 Hypothesis

Based on the research question, we pose the following hypothesis-pair:

- **H1:** Using a smaller input device (smartphone keyboard) results in fewer politeness markers in user utterances compared to using a laptop keyboard.
- **H0:** There is no relationship between input device and the use of politeness markers in user utterances.

6 Data collection

For our study, it was essential to collect conversations. For this, we created a webpage, which takes on the task of querying demographic information of the participant, collecting consent for the study, as well as doing the actual conversations. Before and during the study, we were on a call with the participant, to supervise and clarify any open points.

Before starting, they were informed what device they were assigned for the study, without being given any potentially biasing information about the reasons. Participants were then given a personalized link that took them to their session on the website. There, they were given a consent form, and after agreeing, were forwarded to a questionnaire about their demographics and some other variables, such as age, (optional) student number, prior experience with AI chat systems, cooking experience, frequency of cooking from recipes, general device preference for typing, device preference for longer chat sessions, typing speed in words per minute, comfort with written English, and an optional free-text field for any additional relevant notes. For the typing speed, we then had the participants go to the *monkeytype.com* website and conduct a test to measure their WPM. They were then instructed to enter that value in the form. At this point, the preparation is complete and the website redirects to the tasks.

Unlike the study by Frummet et al. [5], we didn't intend to do a wizard of oz experiment, and instead automated the conversation with an LLM agent. We used OpenAI's gpt5-mini model [13], with a system prompt instructing it to act like a cooking assistant. The model suggests actions to take, asks questions, and encourages a helpful role-play atmosphere to make the experience more immersive and realistic in actual cooking scenarios in the kitchen. The model suggests actions to take, asks questions, and encourages a helpful role-play atmosphere to make the experience more immersive and realistic to actual cooking scenarios in the kitchen.

Every participant was guided through six recipes, the same as with Frummet et al. [5].

Table 1: Recipe overview with number of steps

Recipe	Number of Steps
Parisian Gnocchi	5
Buttermilk-Brined Southern Fried Chicken	6
Duck à l'Orange	10
Savory Cheese Soufflé	8
Pesto alla Genovese	5
Old-Fashioned Apple Pie	6

In order to avoid order-effects, the order in which recipes were presented is randomized. The UI also displays the progress and we ensured that the user can't skip over steps. The model also has safeguards to ensure the user properly participates in the conversation and steers the user if it detects an attempt to skip or deviate from the task. For this, the LLM receives additional context about the user's interactions, keeping track of what step a user is in.

During the conversation, the website collects detailed information about the user's interactions, tracking every input with timestamps, start/completion times for recipes, page transitions, all to

make later analysis easier. The website and the chatbot operated reliably during the study period, with no issues. The collected data was stored in a database, and could be viewed and exported through an admin page.

7 Annotation

In order to annotate the conversations we gathered, we chose to create a python project to process the data using a LLM. Our choice of LLM was OpenAI’s gpt-5-mini-2025-08-07 model, for the high performance, low latency, and affordable cost [13]. Since we estimated we would have to make separate requests per utterance, the cost and speed mattered more than any potential performance improvement in newer models, especially given the limited scope of our ruleset and task.

Before using the model, we had to validate that it produces the same quality of annotations as the human annotators did. For this, we first set up a script harness to ingest the existing data, parsing and joining the annotations and the raw conversations into an usable format. With this, we could now send the individual utterances to the LLM compare them to the expected results.

For our system prompt, we took the detailed explanation of the labeling scheme from Elsweiler et al.[4] and turned it into a set of clearly defined rules. For each utterance, we construct an API request containing the system prompt, as well as a bounded, rolling context window of preceding turns, and the target utterance to be labeled. We decided against using session memory, or sending the whole conversation at once, to ensure reproducible, stable output, and avoid any contamination. This was tested and refined over many steps, including different context windows, and varying instructions. We also tried approaches with few-shot prompting, but none of them delivered measurable improvements, and usually delivered worse overall results. The final code and system prompt are provided in the attached archive.

The high-level workflow of our script ends up being as follows:

- Select one target user utterance.
- Retrieve the previous 4 visible text turns from the same conversation.
- Construct a scheme-aware prompt with explicit annotation instructions.
- Submit target utterance + explicit context to gpt-5-mini.
- Require structured JSON output under a strict schema.
- Parse zero-or-more politeness labels and evidence spans.
- Store one machine-readable record per utterance for downstream analysis.

In the table below, you can see the results we achieved using our model annotation pipeline:

Table 2: Performance of the final selected annotation pipeline on the expert-annotated reference set (user utterances only, $N=677$).

Metric	Score
Exact set match	0.888
Mean Jaccard	0.924
F1: Absolving - ignore	0.800
F1: Absolving – acceptance	0.667
F1: Absolving – denial	0.889
F1: Acknowledge response	0.761
F1: Apology (IFID)	0.000
F1: Compliment/praise/feedback	0.774
F1: Declining	0.500
F1: Hint - statement	0.833
F1: Information-eliciting question	0.951
F1: Politeness marker please	0.966
F1: Request – direct	0.545
F1: Request – indirect	0.897
F1: Request – non-sentential	0.995
F1: Thanking	1.000

As shown, we achieved not only a very high Jaccard score, but also strong F1 scores across the different labels. Although some categories are less frequent in the reference dataset, they were still recognized sufficiently well in manual attempts.

Therefore, we consider those results to be adequate, and used this version of our pipeline on the conversations ($n = 132$) we collected for our study, yielding a total of 1,406 individual utterances. After finishing, we cleaned them up (removing diagnostic information around the LLM API usage etc.) and processed them into a well structured set of csv files, which could then be used for analysis in R Studio. This also included the necessary information about the participants we previously collected.

8 Analysis

8.1 Dataset

Working with the now fully processed dataset, analysis was conducted in R Studio. The ingested data includes 22 participants, with 11 using a phone, and 11 using a desktop. With 6 recipes per participant, we’re looking at 132 conversations, for a total of 1406 utterances.

We can already see that there’s a difference in the number of utterances between mobile and desktop conditions:

Table 3: Annotated user utterances by device type.

Device type	Utterances	Share (%)
Desktop	778	55.3
Mobile	628	44.7

While expected and fitting with our overall assumptions and our hypothesis, the difference is small enough to not affect the validity of our results in a negative way—we have enough utterances for both groups.

8.2 Demographics

Participants' age ranged from 20–28 years ($M = 23.0$, $SD = 2.2$). Their typing speed averaged 54.8 WPM ($SD = 15.2$).

Table 4: Participant age and typing speed by group

Variable	Overall	Desktop	Mobile
Age, M (SD)	23.0 (2.2)	23.9 (2.5)	22.1 (1.4)
Typing speed (WPM), M (SD)	54.8 (15.2)	58.1 (18.1)	51.5 (11.5)

Most participants reported weekly or daily LLM use, and most described themselves as having at least some cooking experience. We can also see a good variation in their previous experiences. We're content with the overall sample, which shows a good variety and demonstrates that our subjects have sufficient context and skills to participate in the study.

Table 5: Participant experience questionnaire.

Variable	Category	n (%)
LLM experience	Few times	4 (18.2)
	Monthly	1 (4.5)
	Weekly	11 (50.0)
	Daily	6 (27.3)
Cooking experience	Beginner	2 (9.1)
	Some experience	10 (45.5)
	Home cook	9 (40.9)
	Very experienced	1 (4.5)
Recipe-following frequency	Rarely	6 (27.3)
	Sometimes	9 (40.9)
	Often	7 (31.8)
English writing comfort	Not very comfortable	0 (0.0)
	Comfortable	12 (54.5)
	Very comfortable	8 (36.4)
	Native / near-native	2 (9.1)

8.3 Descriptive Analysis

Moving on to the actual conversations, we first take a look at them from a quantitative and descriptive perspective.

Table 6 provides an overview of how the annotated politeness-related labels are distributed across the two device conditions, showing the proportion of user utterances in each group that were assigned each label. We can already see some interesting differences between the two conditions—while mobile has a lot more in the *Hint statement* category, we see a lot more diversity in the desktop group. We can also already see that the strongest politeness categories are strongly overrepresented in the desktop group, such as *Thanking* and *Politeness marker please*.

Table 6: Prevalence of politeness-related labels by device type.

Label	Desktop (%)	Mobile (%)
Hint statement	37.1	41.9
Acknowledge response	35.7	31.5
Information-eliciting question	26.2	25.0
Request, non-sentential	9.6	9.6
Request, direct	9.1	5.7
Declining	5.5	2.5
Politeness marker <i>please</i>	4.5	0.5
Compliment / praise / face-enhancing feedback	4.2	4.6
Thanking	3.6	1.3
Request, indirect	3.3	4.9
Apology (IFID)	0.3	0.0

8.4 Typing Speed/Group selection

We now turn to our hypothesis, beginning with the assumption regarding group selection, namely the correlation between typing speed and device form factor.

For that, we look at typing speed between the two groups. Desktop participants have slightly higher typing speeds on average than mobile participants (desktop: $M = 58.1$, $SD = 18.1$; mobile: $M = 51.5$, $SD = 11.5$), although this difference is not *statistically* significant (Welch's $t(17.0) = 1.01$, $p = .326$). However, this still shows that our group selection is directionally correct, and doesn't threaten to work against us.

Next, we examine whether device type is associated with differences in conversational behavior. Desktop participants produce more utterances on average than mobile participants (desktop: $M = 70.7$, $SD = 15.8$; mobile: $M = 57.1$, $SD = 15.6$), with the difference approaching conventional significance (Welch's $t(20.0) = 2.04$, $p = .055$).

We observed a similar pattern for total typed characters (desktop: $M = 2057.3$, $SD = 1001.9$; mobile: $M = 1345.5$, $SD = 509.4$), $t(14.8) = 2.10$, $p = .053$.

By contrast, mean utterance length differences were smaller, both in characters (desktop: $M = 30.4$, mobile: $M = 24.9$, $p = .334$) and in words (desktop: $M = 6.35$, mobile: $M = 5.26$, $p = .340$).

Table 7: Typing speed and conversation measures by device type.

Measure	Desktop (SD)	Mobile (SD)	$t(df)$	p
Typing speed	58.1 (18.1)	51.5 (11.5)	1.01 (17.0)	.326
Total utterances	70.7 (15.8)	57.1 (15.6)	2.04 (20.0)	.055
Utt./assignment	11.8 (2.6)	9.5 (2.6)	2.04 (20.0)	.055
Total characters	2057.3 (1001.9)	1345.5 (509.4)	2.10 (14.8)	.053
Chars/utterance	30.4 (15.0)	24.9 (10.4)	0.99 (17.8)	.334
Words/utterance	6.35 (2.98)	5.26 (2.12)	0.98 (18.0)	.340

Taken together, these comparisons suggest that the more pronounced device-related differences are in *how much participants said overall*, rather than in the average length of *individual utterances*.

8.5 Validating the hypothesis

To test the main hypothesis, we use a politeness-marker metric based on utterances annotated with *please*, thanking, apology, compliment/face-enhancing feedback, or acknowledgment of response. We chose to group these categories together as they are the strongest politeness indicators in the annotation scheme.

At the participant level, desktop users show a higher mean politeness-marker rate than mobile users (desktop: $M = .444$, $SD = .154$; mobile: $M = .327$, $SD = .127$). This comparison is directionally consistent with the hypothesis, although the corresponding Welch test is only marginal ($t(19.3) = 1.96$, $p = .065$).

Table 8: Politeness-marker use by device type.

Measure	Desktop	Mobile
Participant-level marker rate	0.444	0.327
Markers per 100 utterances	52.1	37.0

Because politeness markers are annotated at the utterance level, while device type varies at the participant level, we model politeness-marker use with a mixed-effects Poisson regression rather than treating utterances as independent observations. The outcome is the total number of politeness markers in each participant–recipe conversation. To account for differences in conversation length, we include the log number of utterances as an offset. We also include recipe identity and assignment order as fixed effects, and participant as a random intercept.

The model shows a lower politeness-marker rate for mobile than desktop users (IRR = 0.72, 95% CI [0.52, 1.00], $p = .049$). With the 95% confidence interval, we can see that there’s a relatively small degree of uncertainty around the exact magnitude of the effect.

Table 9: Mixed-effects Poisson model predicting politeness-marker rate.

Predictor	IRR	95% CI	p
Mobile (vs. desktop)	0.72	[0.52, 1.00]	.049

With this, we’ve met the standard p -value of 0.05, and we reject the null hypothesis. We’re looking at a relatively decently sized effect, with the model showing a lower politeness-marker rate for mobile than desktop users. The estimate is 0.72 (95% CI [0.52, 1.00], $p = .049$), which means that, after accounting for conversation length, recipe, assignment order, and repeated observations within participants, mobile users produce politeness markers at about 72% of the desktop rate. Conversely, the model estimates a reduction of roughly 28% in politeness-marker use for the mobile group relative to the desktop group. This result is consistent with our hypothesis that the more constrained input conditions on mobile devices are associated with reduced use of politeness expressions in task-oriented chatbot interaction.

9 Conclusion

With main findings established, we can reflect on the broader implications of the study. The different prompting styles—especially

around their politeness—have a measurable impact on LLM output [4]. Our results support the view that input method also influences politeness in user–LLM interaction. This places a responsibility on the designers and developers of LLM-based applications and devices to consider how interface and input constraints shape user behavior, and how such effects might be mitigated in practice.

One limitation of the study is the relatively small sample size. Interaction style varies substantially across participants, and a larger sample would make it easier to capture this variation more reliably. We’re confident that our assumption and our analysis holds, but it would be interesting to conduct further research that varies the number of tasks and attempts a within-group design, to better capture that difference with even higher confidence. We also had to consider which markers to actually take into account, since they all capture various *aspects* of politeness. Overall, we’re content with the results though, and we feel that the descriptive analysis we performed also supported our hypothesis and assumptions around user behavior and the group differences. More of these can be found in the attached archive that contains our code.

References

- [1] Anyscale. 2024. Understand LLM latency and throughput metrics. <https://www.anyscale.com/docs/llms/performance/latency-throughput>. Accessed: 2025-01-10.
- [2] Apple Inc. 2024. Apple Watch User Guide. <https://support.apple.com/en-gb/guide/watch/apd92a90f882/watchos>. Accessed: 2025-12-07.
- [3] Elias Dritsas, Maria Trigka, Christos Troussas, and Phivos Mylonas. 2025. Multimodal Interaction, Interfaces, and Communication: A Survey. *Multimodal Technologies and Interaction* 9, 1 (2025). doi:10.3390/mti9010006
- [4] Christine Elsweller and David Elsweller. 2020. Cooking Up Politeness in Human–AI Information Seeking Dialogue. *Journal of Examples* (2020).
- [5] Alexander Frummet, Alessandro Spezziorin, David Elsweller, Anton Leuski, and Jeff Dalton. 2024. Cooking with Conversation: Enhancing User Engagement and Learning with a Knowledge-Enhancing Assistant. *ACM Trans. Inf. Syst.* 42, 5, Article 122 (April 2024), 29 pages. doi:10.1145/3649500
- [6] Bill Gates. 2023. AI is about to completely change how you use computers. *GatesNotes* (November 2023). <https://www.gatesnotes.com/ai-agents>
- [7] Jascha Goltermann. 2025. From Commands to Intent: How AI Is Reshaping Our (Digital) World. *Medium (Bootcamp)* (January 2025). <https://medium.com/design-bootcamp/from-commands-to-intent-how-ai-is-reshaping-our-digital-world-29e2d455e5e>
- [8] IBM. 2025. Stanford HAI 2025 AI Index Report: Key Takeaways. *IBM Think* (2025). <https://www.ibm.com/think/news/stanford-hai-2025-ai-index-report>
- [9] Jeong Ho Kim, Lovenoor Aulck, Ornwipa Thamsuwan, Michael C. Bartha, and Peter W. Johnson. 2014. The effect of key size of touch screen virtual keyboards on productivity, usability, and typing biomechanics. *Human Factors* 56, 7 (nov 2014), 1235–1248. doi:10.1177/0018720814531784
- [10] Zijian Lew, Joseph B. Walther, Augustine Pang, and Wonsun Shin. 2018. Interactivity in Online Chat: Conversational Contingency and Response Latency in Computer-mediated Communication. *Journal of Computer-Mediated Communication* 23, 4 (jul 2018), 201–221. doi:10.1093/jcmc/zmy009
- [11] Weixin Liang, Yaohui Zhang, Mihai Codreanu, Jiayu Wang, Hancheng Cao, and James Zou. 2025. The Widespread Adoption of Large Language Model-Assisted Writing Across Society. arXiv:2502.09747 [cs.CL]. <https://arxiv.org/abs/2502.09747>
- [12] Christine Murad, Heloisa Candello, and Cosmin Munteanu. 2023. What’s The Talk on VUI Guidelines? A Meta-Analysis of Guidelines for Voice User Interface Design. In *Proceedings of the 5th International Conference on Conversational User Interfaces* (Eindhoven, Netherlands) (CUI ’23). ACM, New York, NY, USA. doi:10.1145/3571884.3597129
- [13] OpenAI. 2026. *GPT-5 mini Model*. <https://developers.openai.com/api/docs/models/gpt-5-mini> OpenAI API documentation. Accessed: 2026-04-03.
- [14] Johannes Schneider. 2025. Mental model shifts in human–LLM interactions. *Journal of Intelligent Information Systems* 63, 3 (2025). doi:10.1007/s10844-025-00960-6
- [15] ANDREW SEARS, DOREEN REVIS, JANET SWATSKI, ROB CRITTENDEN, and BEN SHNEIDERMAN. 1993. Investigating touchscreen typing: the effect of keyboard size on typing speed. *Behaviour & Information Technology* 12, 1 (1993), 17–22. arXiv:https://doi.org/10.1080/01449299308924362 doi:10.1080/01449299308924362
- [16] Ziqi Yin, Hao Wang, Kaito Horio, Daisuke Kawahara, and Satoshi Sekine. 2024. Should We Respect LLMs? A Cross-Lingual Study on the Influence of Prompt

- Politeness on LLM Performance. (2024). arXiv:2402.14531 [cs.CL] <https://arxiv.org/abs/2402.14531>
- [17] Martina Ziefle. 2010. Information presentation in small screen devices: The trade-off between visual density and menu foresight. *Applied Ergonomics* 41, 6 (2010), 719–730. doi:10.1016/j.apergo.2010.03.001 Special Section: Selection of papers from IEA 2009.